

WASP-11b

R-2020

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_121213 · created 2026-08-08T00:02:39+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456274.73547 +/- 0.00053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.015
Transit depth (Rp/Rs)^2	1.65 +/- 0.39 [%]
Orbital Inclination (inc)	88.41 +/- 0.78
Airmass coefficient 1 (a1)	1.076 +/- 0.01
Airmass coefficient 2 (a2)	-0.063 +/- 0.031
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	None
Optimal Aperture	3.88
Optimal Annulus	4.4
Transit Duration (day)	0.1007 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.128 ± 0.015 vs 0.131 (deviation 0.2σ)
Duration fit vs published	0.1007 d vs 0.1075 d (deviation 1.26σ)
Mid-time O–C	-1.4 min
Depth SNR	8.5
Residual scatter	0.93 %

Light curve

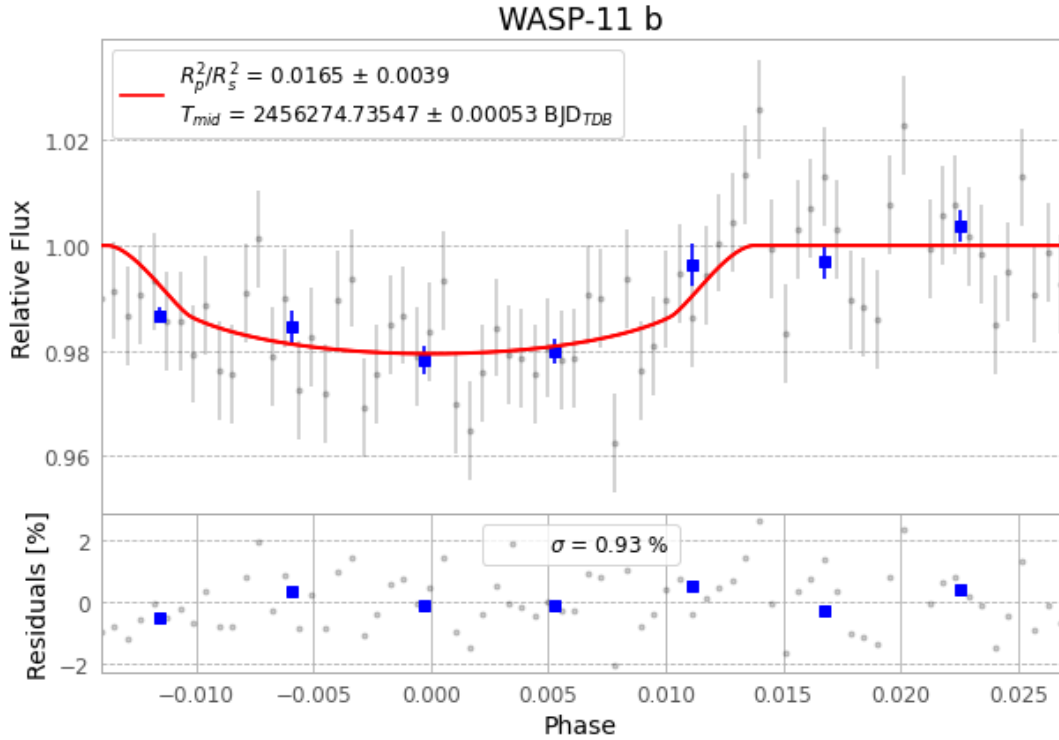


Figure 1 — FinalLightCurve_WASP-11 b_2012-12-12.png (SHA-256 f6ee4eef08e7a7f5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e3baca104bc42ba0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

74 FITS frames (49 MB), HATP-10121213041512.FITS → HATP-10121213075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-121213.log (0b73abc68f16...), FinalLightCurve_WASP-11 b_2012-12-12.png (f6ee4eef08e7...), FinalLightCurve_WASP-11 b_2012-12-12.csv (ae2e0a8a260b...)

Compute resources

Wall time 297.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:02Z.